

Sl No: 4

(Revision – 2021)

Subject Code:

A

Reg. No.

Signature.

Third Semester Diploma Examination in Printing Technology

FUNDAMENTALS OF COLOR REPRODUCTION-SET-I

[Maximum Marks: 75]

[Time: Hours]

PART – A

I. Answer all questions in one word or one sentence.

- 1 Recall color sensitive nerve cells in the human eye.
- 2 Define Tristimulus values.
- 3 Recall WYSIWYG principle.
- 4 Specify the formula to calculate Print Contrast.
- 5 Show the schematic sketch of additive color theory model.
- 6 Define color gamut.
- 7 Specify the peculiarities of light sources used in color measuring devices.
- 8 Recall any four halftone dot shapes.
- 9 Specify the major tonal gradations of a continuous tone image.

(9 x 1 = 9 Marks)

PART B

II. Answer any *eight* questions. Each question carries 3 marks.

1. Relate color and Light.
2. Recall the peculiarities of CIE Lab Color Space.
3. Spell the advantages of Color management.
4. Calculate Delta E value of two printed samples, if $L_1=72$, $a_1=+85$, $b_1=-34$, $L_2=55$, $a_2=+92$, $b_2=-20$.
5. Tell about CIE milestones related to color and light.
6. Specify the features of monochromatic color schemes including its use.
7. Recall 3 C's of color management.
8. Examine the role of independent color model in multicolor printing.
9. Summarize the significance of halftone dots in printing.
10. If $D_1=1.5$, $D_2=1.0$, $D_1+2=1.75$, calculate the ink trapping and declare its acceptance.

(8 x 3 = 24 Marks)

PART C

Answer all questions. Each question carries 7 marks.

III. Define Metamerism and explain the relevance of its existence in printing.

OR

IV. Differentiate Additive and Subtractive color theories.

V. Illustrate and explain CIE chromaticity diagram.

OR

VI. Explain about CIE Lab color space with necessary sketch.

VII. Explain the working principle of spectrophotometer.

OR

VIII. Describe the functions of CMM, PCS & Color rendering index of CMS.

IX. Estimate the dot gain of given situation, if a measurement of 50% dot area is obtained from the film and a 67% dot area is obtained from a proof or press sheet.

OR

X. Estimate the Hue Error and Grayness of a print with density values-H=1.18, M=0.48, L=0.22.

XI. Explain about color separation by scanner .

OR

XII. Explain the importance and need of reproducing tonal gradation in printing.

XIII. Correlate Tristimulus values with Chromaticity Diagram.

OR

XIV. CIE color matching experiment 1931.

(6x 7 = 42 Marks)

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Third Semester Diploma Examination in Printing Technology

FUNDAMENTALS OF COLOR REPRODUCTION-SET-II

[Maximum Marks: 75]

[Time: Hours]

PART – A

I. Answer all questions in one word or one sentence.

- 1 Recall any two memory colors.
- 2 Expand CIE.
- 3 The formula of chromaticity coordinate, $x = \dots\dots\dots$ & $y = \dots\dots\dots$
- 4 Define color model.
- 5 Device profile means.....
- 6 Specify elements involved in color measurement process.
- 7 Identify the shape of given halftone dot. 
- 8 $\Delta E = (\dots\dots\dots)^{1/2}$
- 9 Recall any one reason of obtaining metamerism in print.

(9 x 1 = 9 Marks)

PART B

II. Answer any *eight* questions. Each question carries 3 marks.

11. Illustrate color separation performed in a PMT scanner.
12. Recall munsell color model.
13. Tell about the color conversion process performed by a CMS.
14. Specify the features of tristimulus color measurement.
15. If $L_1=80, a_1=120, b_1=-80, L_2=75, a_2=100, b_2=-20$, Calculate ΔE .
16. Tell about the role of warm & cool color schemes in designs.
17. Define color temperature and spell its importance.
18. Summarize the importance of color management in printing.
19. Examine the influence of grey-level in printing.
20. Define dot gain and list out the reasons of mechanical dot gain.

(8 x 3 = 24 Marks)

PART C

Answer all questions. Each question carries 7 marks.

III. With a neat sketch describe about spectral power distribution curve and its importance.

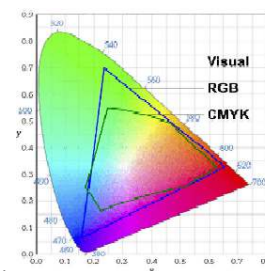
OR

IV. With a neat sketch of elements of eye explain the color perception process of human eye.

V. Explain the background of Colorimetric concepts based on CIE color matching experiment of 1931.

OR

VI.



Compare the color gamut of given systems and analyze the reasons.

VII. With an illustration, explain about the basic working principle of tristimulus color measurement by a colorimeter.

OR

VIII. With sketch of a Digital workflow explain about the color transformation take place in image reproduction process.

IX. Perform Dot area measurement formula by Murray Davis equation. Take Solid patch Density $C=1.75$, $M=1.8$, $Y=1.7$, $K=1.5$ as and Tint patch density as $C=0.75$, $M=0.80$, $Y=0.7$, $K=0.5$.

X.

OR

A pressman may view a halftone print under a loupe and determine that it is a 50% dot area as that of same experienced with the plate image. But when that same halftone is viewed by the naked eye from typical reading distance it appears as 70% dot area. Calculate the Mechanical and optical dot gain.

XI. Explain the Electronic principles of color separation.

OR

XII. Explain about Maxwell Triangle.

XIII. Illustrate CMS workflow and explain its role in ink optimization.

OR

XIV. Explain the working of PCS and Rendering intents.

(6 x 7 = 42 Marks)

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